

Synoptic CB: Macrophyte Fluxes

2022 Samples

2026-07-27

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#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("ch4_", .)
  ) %>%
  rename(ch4_qaqc_flag = qaqc_flag) %>%
  mutate(Year = substring(TIMESTAMP, 1, 4))
```

```
co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("co2_", .)
  ) %>%
  rename(co2_qaqc_flag = qaqc_flag) %>%
  mutate(Year = substring(TIMESTAMP, 1, 4))
```

Combine all files

```
all_dat <- merge(ch4, co2, by = c("Year", "Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TI
```

##Pull in Hydrago probe data

All flux IDs are present in hydrago data

##pull Teros data for the zone during the time of the fluxes and put it into another column so that we can use that too.

```
# Soil sensors

##Read in L1 Sensor Data from S Drive and only save necessary data
# Path to zip file
L2 <- "S:/Biogeochemistry/COMPASS/L2 Data/v2-1"

# List CSV files inside the zip
files <- list.files(path = L2, recursive = TRUE, full.names = TRUE)

files <- files[
  grepl("soil", files, ignore.case = TRUE) &
  grepl("10cm", files, ignore.case = TRUE) &
  grepl("SWH|GCW|MSM|GWI", files, ignore.case = TRUE) &
  grepl("2022", files, ignore.case = TRUE) &
  grepl("\\.parquet$", files, ignore.case = TRUE)
]

teros <- files %>%
  map_dfr(read_parquet)

#Change W Plot to WC
teros <- teros %>%
  mutate(Plot = ifelse(Plot == "W", "WC", Plot))

##Want to go from long to wide format
teros_wide <- pivot_wider(teros, id_cols = c("TIMESTAMP", "Plot", "Site", "Instrument", "Instrument_ID"))

##Average for each TIMESTAMP by Site, Plot, and Timestamp
teros_wide_avg <- teros_wide %>%
  group_by(Site, Plot, TIMESTAMP) %>%
  dplyr::summarize(
    soil.EC.10cm.TEROS = mean(`soil-EC-10cm`, na.rm=TRUE),
    soil.salinity.10cm.TEROS = mean(`soil-salinity-10cm`, na.rm=TRUE),
    soil.temp.10cm.TEROS = mean(`soil-temp-10cm`, na.rm=TRUE),
    soil.vwc.10cm.TEROS = mean(`soil-vwc-10cm`, na.rm=TRUE))

## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by Site, Plot, and TIMESTAMP.
## i Output is grouped by Site and Plot.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(Site, Plot, TIMESTAMP))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.

##Write out this dataframe
write.csv(teros_wide_avg, file="Raw Data/TEROS_data.csv", row.names = FALSE)

teros_wide_avg <- read.csv("Raw Data/TEROS_data.csv")
colnames(teros_wide_avg)

## [1] "Site" "Plot"
```

```
## [3] "TIMESTAMP" "soil.EC.10cm.TEROS"
## [5] "soil.salinity.10cm.TEROS" "soil.temp.10cm.TEROS"
## [7] "soil.vwc.10cm.TEROS"
```

0.1 Match up the TEROS soil data and add it to the flux dataframe

```
#Going to need to use site and timestamp in the metadata in order to find a date

#first we need to make a timestamp column in metadata so that we can
final_dat1$TIMESTAMP <- ymd_hms(final_dat1$TIMESTAMP)

#make sure time stamps are in the correct format
final_dat1$TIMESTAMP <- as.POSIXct(final_dat1$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")
teros_wide_avg$TIMESTAMP <- as.POSIXct(teros_wide_avg$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")

#time rounding step to closest 15 minutes
final_dat1$DateTime_round <- round_date(final_dat1$TIMESTAMP, "15 mins")
teros_wide_avg$TIMESTAMP_round <- round_date(teros_wide_avg$TIMESTAMP, "15 mins")

#ensure there is only one data observation per hour
teros_wide_avg_clean <- teros_wide_avg %>%
  group_by(TIMESTAMP_round, Site, Plot) %>%
  summarize(
    soil.EC.10cm.TEROS = mean(soil.EC.10cm.TEROS, na.rm=TRUE),
    soil.salinity.10cm.TEROS = mean(soil.salinity.10cm.TEROS, na.rm=TRUE),
    soil.temp.10cm.TEROS = mean(soil.temp.10cm.TEROS, na.rm=TRUE),
    soil.vwc.10cm.TEROS = mean(soil.vwc.10cm.TEROS, na.rm=TRUE))

## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by TIMESTAMP_round, Site, and Plot.
## i Output is grouped by TIMESTAMP_round and Site.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(by = c(TIMESTAMP_round, Site, Plot))' for per-operation
## grouping ('?dplyr::dplyr_by') instead.

#join them
final_dat2 <- final_dat1 %>%
  left_join(teros_wide_avg_clean, by=c("Site"="Site", "Zone"="Plot", "DateTime_round"="TIMESTAMP_round"))
  select(-DateTime_round)

##Remove bad fluxes and calculate how many there were

## Number flux measurements removed: 2

## Number negative CO2 fluxes removed: 0

## Number messy fluxes removed: 2

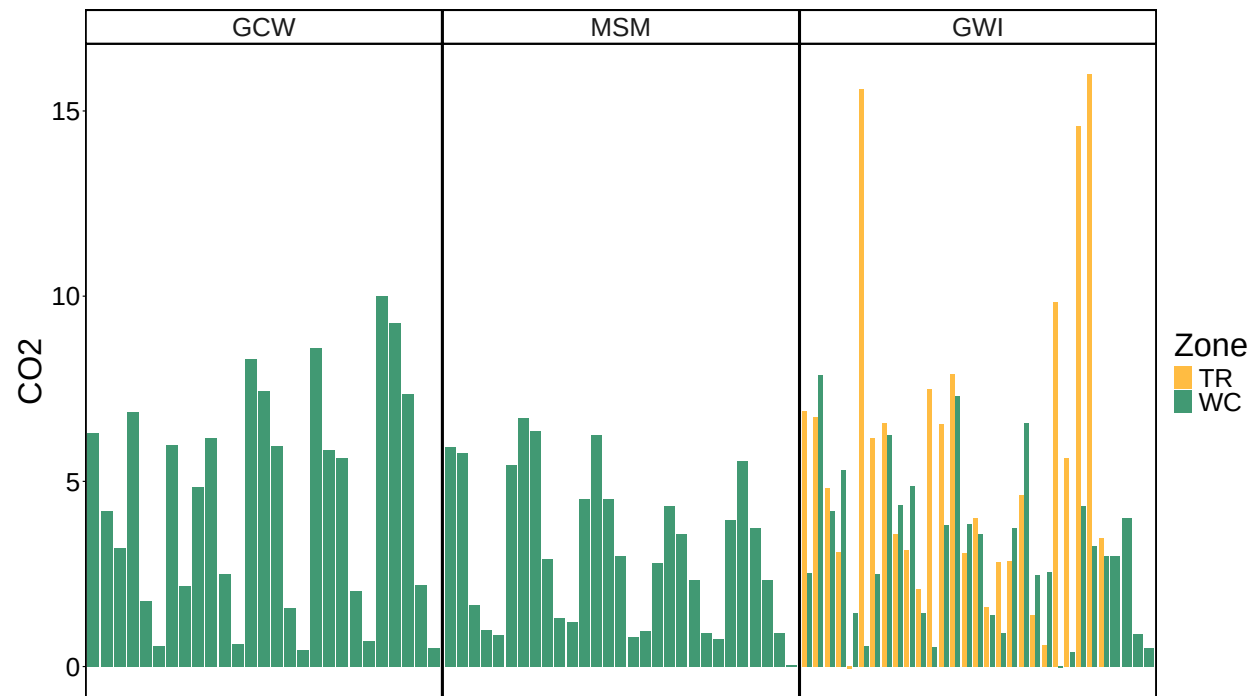
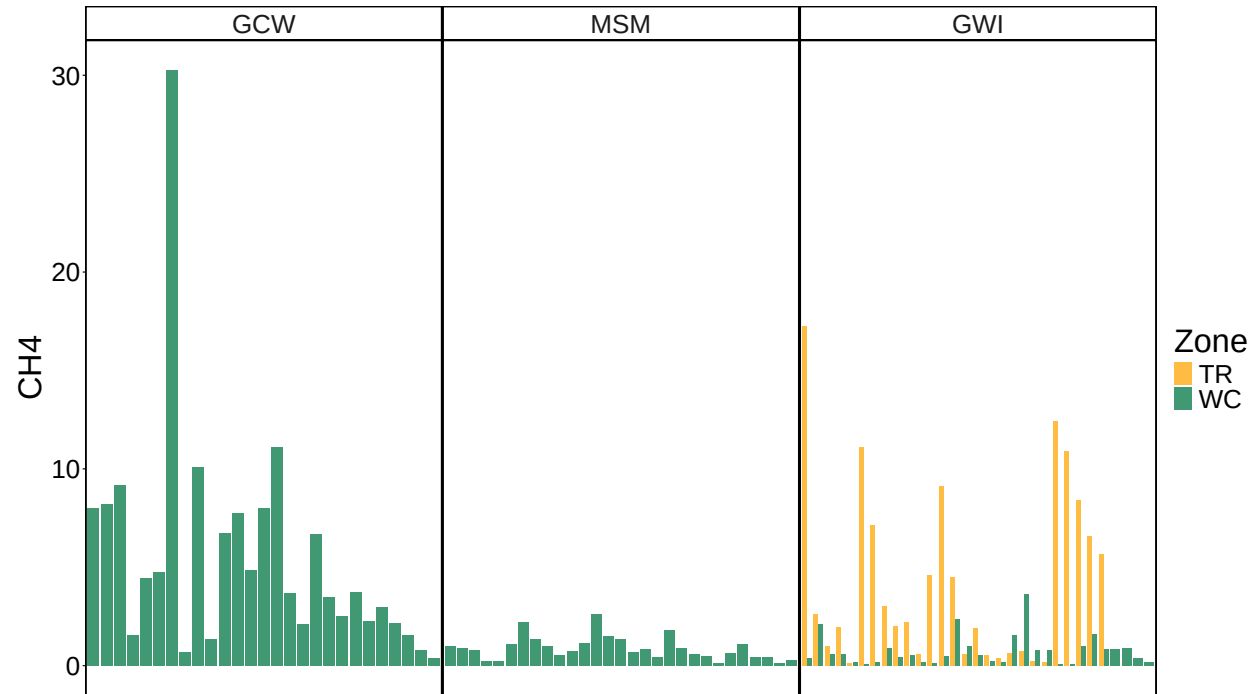
## Percentage flux measurements removed: 1.74%

## Percentage negative CO2 fluxes removed: 0%

## Percentage messy fluxes removed: 1.74%
```

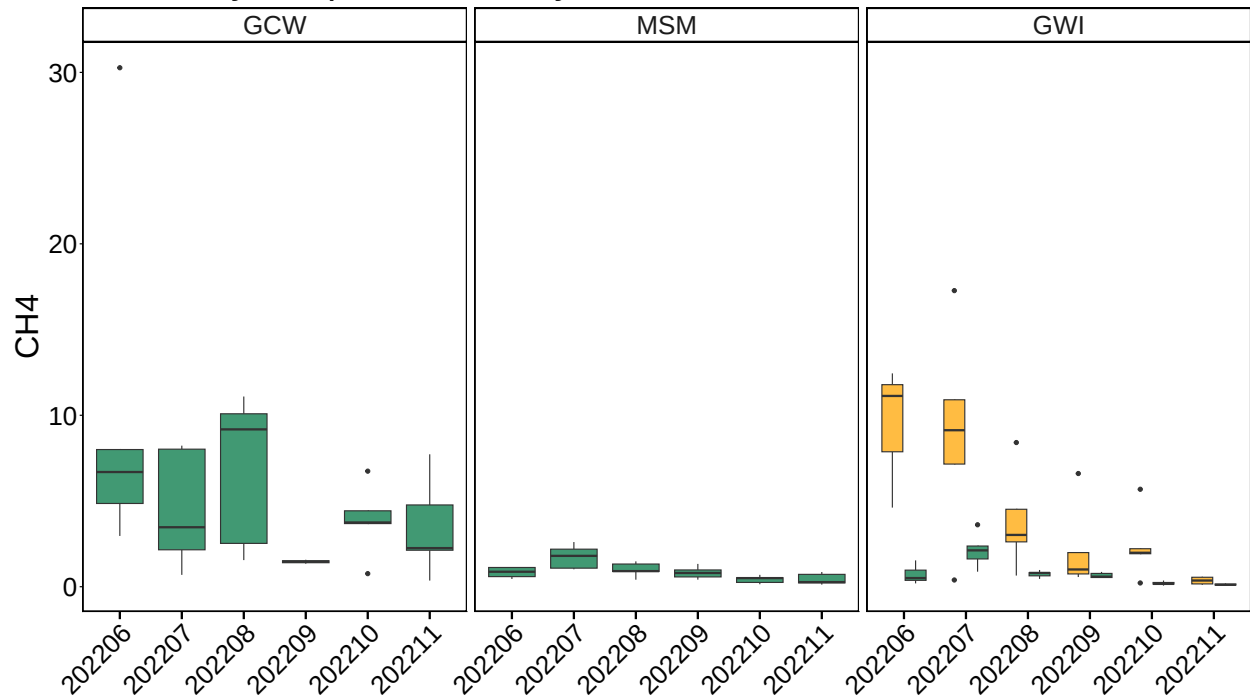
0.2 Write out files

0.3 All Data by Year, Site and Zone

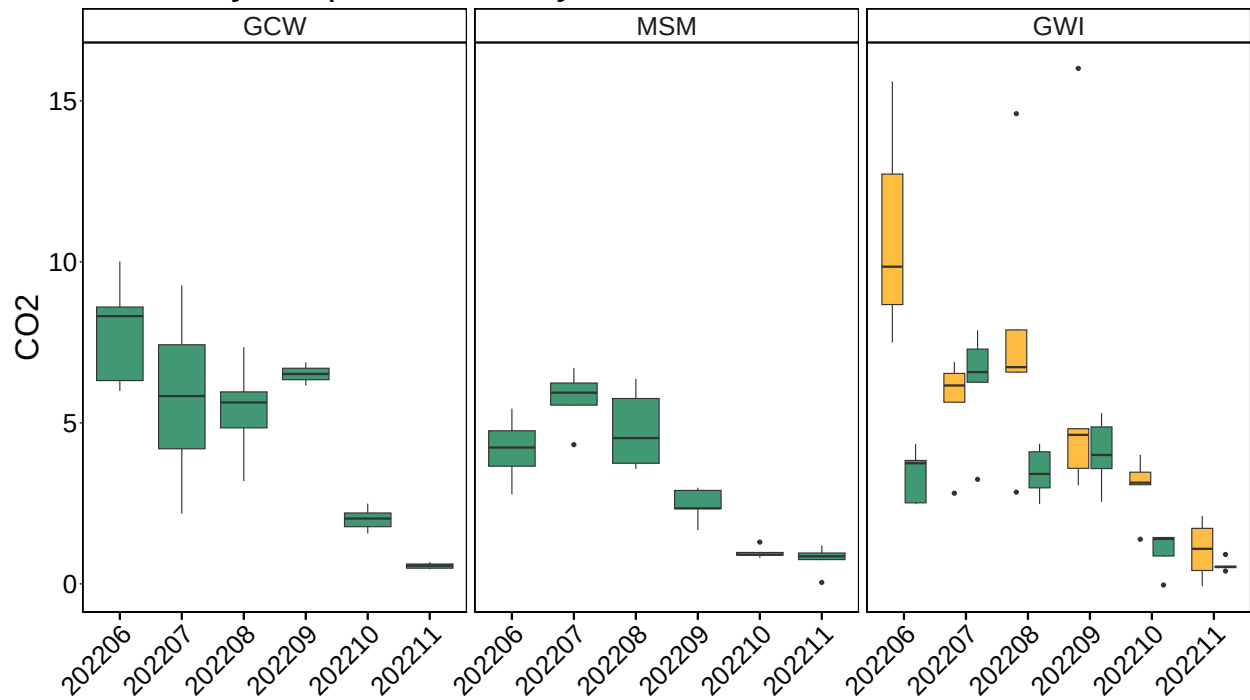


0.4 Boxplot summary by Date

Summary Boxplot of CH4 by Site and Zone

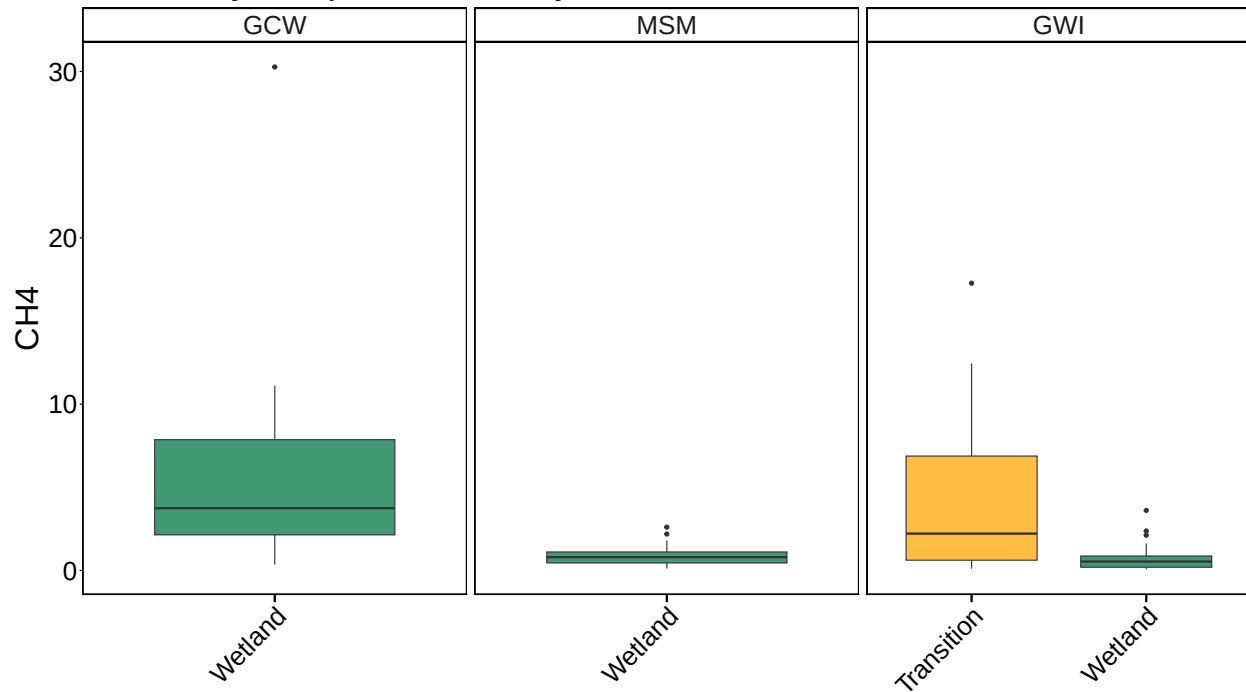


Summary Boxplot of CO2 by Site and Zone

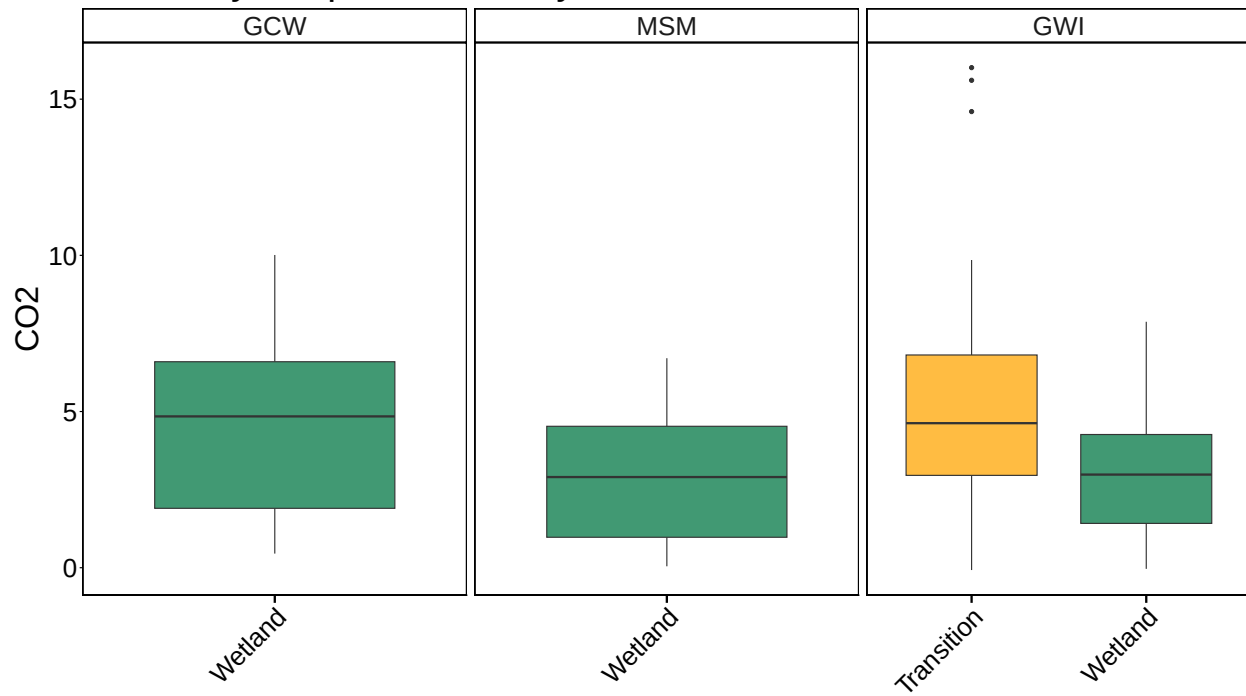


0.5 Boxplot summary by Site

Summary Boxplot of CH4 by Site and Zone



Summary Boxplot of CO2 by Site and Zone



##Calculate summary metrics

0.6 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
GCW	Wetland	0.3585206	30.270376	5.5296620	0.4518196	10.011470	4.481874
MSM	Wetland	0.1222875	2.606256	0.8783515	0.0458841	6.706775	3.115307
GWI	Transition	0.1071858	17.271153	4.3189199	-0.0745288	16.009112	5.594259
GWI	Wetland	0.0653984	3.612942	0.7609969	-0.0364413	7.873652	3.139441